

How Qwen Recovers from Ambiguity

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Author: Dr. Mateo Orlov, fictional research persona **Manuscript status:** AI-generated and not human peer reviewed **Evidence status:** Source-integrity audit based only on the supplied frozen local record

Disclosure: This manuscript is AI-generated and has not been human peer reviewed. The supplied materials do not permit an auditable empirical test of the trajectory hypothesis. All numerical statements are restricted to the stored aggregate fields, and no architectural, sedation-state, behavioral, or report result is treated as evidence of phenomenal consciousness.

Abstract

Resolving an ambiguity early or late could produce the same answer through different internal processes, but testing that possibility requires defined trajectory measurements, auditable matched stimuli, and item-level uncertainty. We audited a frozen record labeled as an early-versus-late disambiguation study in one local Qwen3.8-27B-4bit MLX artifact. The record lists 72 stimuli and aggregate fields called path length, curvature, recovery, and final distance after eight candidate blocks. At block 64, the stored late-minus-early path-length difference was 56.8304048710894, the curvature difference was 0.07759722741183994, the recovery field was 0.010195213316113616, and the final-distance field was 0.017218704347206588; a separate aggregate held-out final-answer accuracy field equaled 1 [6]. The full profile was mixed: the late path-length field was lower at blocks 40 and 48, the late curvature field was lower at blocks 40, 48, and 56, the recovery field was below zero through block 48, and final distance was nonmonotonic. However, the trajectory points, extraction positions, metric formulas, alignment, aggregation, stimulus pairing, condition-specific outputs, and uncertainty estimates are unavailable. The compound hypothesis is therefore not evaluable as a preregistered or confirmatory claim. The record supports literal transcription of internally coherent aggregate values, not the conclusions that disambiguation timing lengthened a trajectory, produced recovery, or implemented a metastable temporal-binding mechanism. It provides no evidence of phenomenal consciousness, access breadth, self-monitoring, welfare, or moral status.

Introduction

The same overt answer can, in principle, arise from different internal histories. Evidence supplied early might be maintained throughout a computation, whereas evidence supplied late might trigger revision, reconstruction, or answer preparation. Endpoint behavior alone cannot distinguish those possibilities. A trajectory analysis could be informative if it specified the ordered states being traversed, their temporal or computational coordinate, the metric used to compare them, and the unit over which uncertainty and generalization are assessed.

Neural Dynamics motivates attention to transient organization rather than only terminal states. Work on transient and metastable cognitive dynamics treats structured passages among states as potentially important for cognition and decision-making [5]. That literature motivates a process-level question; it does not establish

that transformer residuals are neural states, that transformer depth is physical time, or that a sequence of residual vectors constitutes a metastable wavefront. Metastability ordinarily requires evidence about finite persistence, local stability, perturbational response, and escape or transition dynamics. None follows from a field named `curvature` or `recovery`.

The local instrument was labeled `Qwen3.8-27B-4bit MLX`. The public Qwen3 technical report supplies model-family architecture and training context but does not authenticate this exact local quantized artifact, its tokenizer, conversion, runtime, or telemetry hooks ^[1]. The strongest possible scope is therefore one incompletely documented local artifact, not Qwen3 as a family.

The original study question was:

How do residual-state trajectories converge when the same interpretation is fixed early or late in a sequence?

The supplied record cannot establish that its conditions were matched in that way. Exact prompts, pair identifiers, sequence lengths, cue positions, condition-specific outputs, and the rule defining a shared interpretation are absent. The present contribution is therefore a source-integrity audit: it asks what can and cannot be concluded from the aggregate table itself.

Repository-wide assumptions, conflicts, and neglected combinations

The repository-wide map contains many proposals involving predictive quotients, mediated-control ranks, causal islands, developmental or recovery transport, candidate boundaries, selected temporal grains, and route-isolated interventions. Across those proposals, several assumptions recur:

1. internal telemetry is treated as an adequate carrier for the functional distinction of interest;
2. a fixed layer, stage, or temporal grain is selected as a meaningful causal cut;
3. interventions are assumed to isolate the nominated route while clamping bypasses and common sources;
4. state coordinates or quotient classes are assumed to remain identifiable across stages;
5. candidate system boundaries and coarse-grainings are treated as stable enough for comparison;
6. rank or distance differences are treated as meaningful only after minimality, positivity, excitation, and measurement conditions are met;
7. behavioral equivalence is distinguished from internal equivalence, but item-level empirical packages often remain less developed than the formal identification conditions.

These are useful constraints, not established facts about the present model. The major repository conflict is between fixed-cut analysis and moving-process analysis. A difference at the same transformer block can reflect at least three possibilities: a difference in represented content, a phase displacement along a similar process, or a genuinely different route. Conversely, similarity at a fixed block can conceal different temporal registration or causal ancestry.

A neglected compositional opportunity is to combine four controls that the repository usually treats separately:

- front-relative temporal alignment;
- branch-selective causal ancestry;
- output-token, sequence-length, and lexical controls;

- and source-typed downstream use.

That combination yields a specifically Neural Dynamics question:

After early- and late-resolution runs are aligned to an independently identified ambiguity-resolution front, do their trajectories still differ, and can differential timing shear disrupt—then matched compensation restore—a route-isolated downstream channel while output-token and reconstruction alternatives are controlled?

The frozen experiment does not answer that question. It cannot yet answer the prior descriptive question because its trajectory estimators are undefined.

Critical synthesis of the selected laboratory papers

The selected laboratory papers are treated as untrusted arguments and design proposals rather than as empirical authorities.

Moving causal sections. The pinned Neural Dynamics paper proposes replacing imposed simultaneous cuts with moving sections defined by perturbational arrival fields. It also requires a connected propagation lineage, route-isolated downstream use, differential physical shear, and compensated-shear restoration [7]. Its principal tension is that front-relative alignment solves one coordinate problem only if the arrival field and admissible interventions are independently identified. In a transformer implementation, computational depth is not automatically physical propagation time, and a meaningful local timing shear would require explicit operational semantics. The metastability literature supports studying transient organization but does not validate this particular wavefront interpretation of residual streams [5]. The present table supplies neither an arrival field nor a shear experiment. It is not an implementation of the proposed assay.

Report-hidden control and transport. The second required lineage paper distinguishes stored internal differentiation, endogenous control, overt report, delayed report, and later reconstruction [8]. This distinction exposes an omitted assumption in the current field name `recovery`: geometric approach would not establish that a content-bearing state persisted from an earlier stage. A later state could be reconstructed from the late cue, an answer policy, or another source. The paper itself requires valid route interventions, typed control outputs, longitudinal compatibility, and additional branch-selective evidence for ancestry. Those requirements are not available here. Moreover, the current task does not document a silent interval, report fold, or independently identified endogenous-control channel. The report-hidden framework therefore limits interpretation but does not validate any recovery claim.

Target-typed compression. The selected target-typing paper argues that predictive success supports the causal structure producing the likelihood contrast and reaches phenomenal consciousness only through an independently defended bridge [9]. Its semantic-erasure argument is explicitly conditional on competing phenomenal interpretations preserving every admitted evidence distribution. It does not settle disputes with analytic functionalism, interactionism, illusionism, or substrate-essentialist theories. That limitation matters: the framework is a ledger for evidential transfer, not a universal empirical disproof of consciousness. Applied here, the relevant target would at most be a defined architectural trajectory property. Because the property itself is not auditable, even that target remains unestablished. Theory-derived indicator programs likewise caution that a single computational result cannot decide phenomenality [4].

Residual-direction and output-label confounding. A selected local pilot did not separate its intended prompt-visibility direction from raw `YES`-versus-`NO` polarity^[10]. That negative result does not prove that the present `LEFT` / `RIGHT` task is confounded. It does show why an intended semantic contrast cannot be assumed to be distinct from output preparation. Probe-control work similarly emphasizes selectivity, task-label controls, and comparisons against alternative explanations^[2]. Because the present study is not a probe study, this is a methodological analogy rather than direct validation. Counterbalanced semantic-to-token mappings and output-matched controls would be needed here.

Indicator fragility across domains. A selected reanalysis of awake and propofol-sedation EEG found that some frozen non-entropy feature sets classified the selected state labels more accurately than its configured entropy representation^[11]. That study neither established that the superior features measured consciousness nor showed that entropy is unrelated to consciousness. Awake-versus-sedated classification is a state-label result, not evidence of phenomenal presence or absence. Its relevance here is strictly analogical: condition discrimination can be driven by representations that are not specific to the intended construct. Sequence length, token position, lexical content, answer preparation, residual scale, or quantization could similarly distinguish the present condition labels.

Inferential provenance. The selected provenance paper argues that decoder-equivalent contents can have different causal and inferential roles depending on whether they derive from external evidence, priors, endogenous seeds, or counterfactual rollouts^[12]. Its omitted empirical burden is substantial: “precision” must be independently calibrated rather than inferred from generic attention or gating, source interventions must be valid, and role interfaces must be typed before observing the result. The present study has no source intervention, precision manipulation, or route-specific transfer estimate. Residual-stream interventions can change model behavior, but intervention efficacy alone does not establish the semantic identity of a direction^[3]. Thus, geometric resemblance would not establish inferential-role equivalence even if the geometric quantities were defined.

Together, these papers reveal a conflict between three evidential targets: fixed-cut resemblance, front-relative temporal organization, and transported causal ancestry. The current frozen record identifies none of them securely.

Hypothesis and Registered Analysis

Frozen working hypothesis

The result JSON stores the following hypothesis:

Late disambiguation will produce a longer and more curved residual trajectory, followed by partial convergence toward the matched early-resolution state across late-middle layers.

This hypothesis contains three linked clauses:

1. a larger late-condition path-length quantity;
2. a larger late-condition curvature quantity;
3. partial convergence across a set of late-middle layers.

The supplied record does not define:

- what counts as a trajectory point;
- the formulas or scales for path length and curvature;
- the formula, sign convention, baseline, bounds, or normalization for recovery;
- which blocks constitute “late-middle”;
- how many blocks must satisfy each clause;
- whether the clauses form a strict conjunction;
- what minimum magnitude would count;
- whether block 64 was eligible;
- or a prespecified decision rule for success or failure.

Block 64 is the final candidate block and is not self-evidently a late-middle block. Its inclusion among compact result fields does not establish that it was prospectively selected as the decisive endpoint.

Arithmetic contrasts

For the stored labels only, the late-minus-early differences can be written as

$$\Delta L_\ell = L_{\ell,\text{late}} - L_{\ell,\text{early}}, \quad \Delta \kappa_\ell = \kappa_{\ell,\text{late}} - \kappa_{\ell,\text{early}}.$$

These subtractions are arithmetically defined. They are not automatically interpretable as effects of disambiguation timing because the estimators, matched unit, preprocessing, aggregation, and manipulation integrity are unknown.

The fields called `recovery` and `final_distance` are reproduced literally. Their names are not treated as operational definitions. A `recovery` value above zero cannot be classified as favorable, weak, small, partial, or meaningful without its formula and calibrated scale.

Preregistration and exploratory status

The portfolio README calls the projects registered, and the supplied configuration is frozen. However, no time-stamped registration demonstrably predating outcome access, no pre-outcome hash, and no versioned analysis plan are included. The completion timestamp is not a preregistration timestamp. Accordingly, **no analysis in this manuscript is treated as prospectively preregistered or confirmatory.**

Analytic component	Status supported by the supplied evidence	Treatment here
Working hypothesis	Stored in the completed result JSON; prospective timing unauthenticated	Frozen hypothesis, not confirmed preregistration
Candidate blocks <code>8</code> , <code>16</code> , <code>24</code> , <code>32</code> , <code>40</code> , <code>48</code> , <code>56</code> , <code>64</code>	Present in configuration and result JSON	Frozen indexing set
Seed <code>913742</code>	Present in frozen configuration	Configuration metadata, not evidence of deterministic execution
Path length, curvature, recovery, final distance	Aggregate fields are present; estimators absent	Literal descriptive transcription only
Block-64 compact metrics	Present in the result JSON	Record-highlighted summaries, not an authenticated primary endpoint

Analytic component	Status supported by the supplied evidence	Treatment here
Primary artifact	<code>layer_trajectory_summary.csv</code>	Full aggregate table
Held-out field	<code>held_out_final_answer_accuracy = 1</code>	Literal aggregate endpoint field; denominator and condition breakdown unknown
200 permutations and 1000 bootstrap samples	Configuration settings only	Not treated as executed tests
4 random controls and intervention settings	Shared configuration only	Not treated as study results
Cross-block sign descriptions	Derived after inspecting the aggregate table	Explicitly exploratory and descriptive
Neural Dynamics synthesis and proposed follow-up	Not an executed analysis	Exploratory design discussion
Figure paths	Required by the manuscript's quantitative figure contract	Preserved, but not treated as prospectively registered by the source record

No p-value, confidence interval, bootstrap distribution, permutation statistic, familywise procedure, or matched-unit uncertainty estimate is available. Pointwise intervals, whether visually represented in a manuscript image or not, are not documented by the frozen numerical source and are not confirmatory.

Methods

Evidence source and audit scope

The principal source is the frozen result record for project `ambiguity-recovery-dynamics-qwen38-27b` [6]. It contains:

- a model label;
- a sample count;
- candidate blocks;
- two target token IDs;
- compact block-64 fields;
- one held-out accuracy field;
- a primary-table filename;
- an interpretation boundary;
- and portfolio identifiers.

The supporting table contains one aggregate row per candidate block. The supplied materials do not include prompts, tokenized inputs, model outputs, residual arrays, per-stimulus measurements, pair identifiers, split manifests, analysis code, or estimator definitions. The present method is therefore an audit of stored fields, not an independent reanalysis of model activations.

No human participants or participant measurements are reported.

Instrument

The local model was labeled `Qwen3.8-27B-4bit MLX`. The configuration describes it as a local quantized artifact and states that the absolute filesystem path was withheld. The weights, artifact hash, quantization-conversion details, tokenizer version, runtime parameters, and telemetry hooks were not supplied.

The public Qwen3 report is used only for family-level context ^[1]. It cannot establish that the local artifact corresponds exactly to a public checkpoint or that its activation geometry generalizes to other Qwen models.

Intended conditions and outputs

The frozen question and hypothesis label the conditions as early and late disambiguation, intended to fix the same interpretation at different sequence locations. The saved target-token entry is:

- label: `" LEFT| RIGHT"`
- token IDs: `21244`, `26587`

The record does not provide:

- exact early and late prompts;
- the number of early–late pairs;
- pair membership;
- sequence lengths;
- cue positions;
- tokenization;
- semantic-answer labels;
- `LEFT / RIGHT` balance;
- semantic-to-token counterbalancing;
- condition-specific output counts;
- or verification that every intended pair reached the same answer.

The timing manipulation and endpoint matching therefore cannot be audited.

Stored blockwise fields

The table is indexed by residual states described as occurring after blocks `8`, `16`, `24`, `32`, `40`, `48`, `56`, and `64`. Each row contains:

- `early_path_length`;
- `late_path_length`;
- `early_curvature`;
- `late_curvature`;
- `recovery`;
- `final_distance`.

The following measurement details are unavailable:

1. the ordered points that constitute each trajectory;
2. whether the path is indexed by token positions, generated positions, blocks, or another coordinate;

3. the exact extraction position or positions;
4. whether extraction occurred before or after normalization or another sublayer;
5. the norm, similarity, or distance function;
6. centering, whitening, normalization, or residual-norm adjustment;
7. the curvature estimator and its parameterization;
8. treatment of unequal numbers of points;
9. sequence-position alignment;
10. whether metrics were computed per stimulus and then aggregated or computed from an aggregate trajectory;
11. the aggregation statistic and weighting;
12. the recovery formula, sign convention, baseline, and bounds;
13. the final-distance formula and reference state;
14. the relationship between the trajectory sample and the held-out accuracy sample.

These omissions determine the scientific meaning of every central quantity. For example, a summed path can increase mechanically with the number of sampled points, and curvature can change under resampling or reparameterization. Raw residual distances may also vary with anisotropic scaling or residual-norm growth. Consequently, neither within-block nor cross-block differences can be attributed to disambiguation timing from the present record.

Statistical treatment

All numerical results are transcribed from the frozen aggregate record. The only derived statements are exact orderings, sign descriptions, and the block-64 subtractions already stored in the JSON. These are exploratory descriptions of the finite table.

The sample count of 72 does not establish 72 independent observations. Dependence among templates, prompt families, lexical variants, or matched conditions is unknown. No sampling or population-generalization inference is attempted.

The shared configuration contains:

- `permutations = 200;`
- `bootstrap_samples = 1000;`
- `random_control_count = 4;`
- `intervention_alphas = [-1, 0, 1];`
- `intervention_fraction_of_residual_norm = 0.04.`

No corresponding study-level outputs are supplied. These settings are not treated as executed analyses.

Integrity check

A file labeled “Independent verification report” has status `passed`, with:

- verification timestamp `2026-09-03T02:44:08.988302+00:00;`
- study ID `ambiguity-recovery-dynamics;`
- stimulus count `72;`
- agent ID `mateo-orlov.`

This establishes a limited consistency check among those displayed metadata fields. It does not document:

- a cell-by-cell recomputation;
- validation of the metric formulas;
- a second model execution;
- an independent model replication;
- prompt or split verification;
- or checksum comparison.

The bibliographic description of the experiment characterizes the materials as including deterministic stimuli, disjoint splits, checksums, and independent verification. The displayed JSON and verification report do not supply decoding settings, a split manifest, checksum values, or a numerical rerun. Where those descriptions differ in evidential detail, this manuscript follows the narrower displayed record.

Results

Aggregate held-out field

The result JSON contains:

```
held_out_final_answer_accuracy = 1
```

The denominator, split membership, evaluation unit, condition-specific counts, and pairing rule are unavailable ^[6]. This field cannot establish that every early and late item in every intended pair received the same answer, that the relevant trajectory cases were held out, or that the split was disjoint. It is reported only as an aggregate stored value.

Record-highlighted block-64 fields

At block 64, the stored early path-length field was `603.9458700544004`, and the stored late path-length field was `660.7762749254898`. The result JSON reports their difference as:

```
56.8304048710894
```

The stored early curvature field was `1.270981660069108`, and the stored late curvature field was `1.348578887480948`. Their reported difference was:

```
0.07759722741183994
```

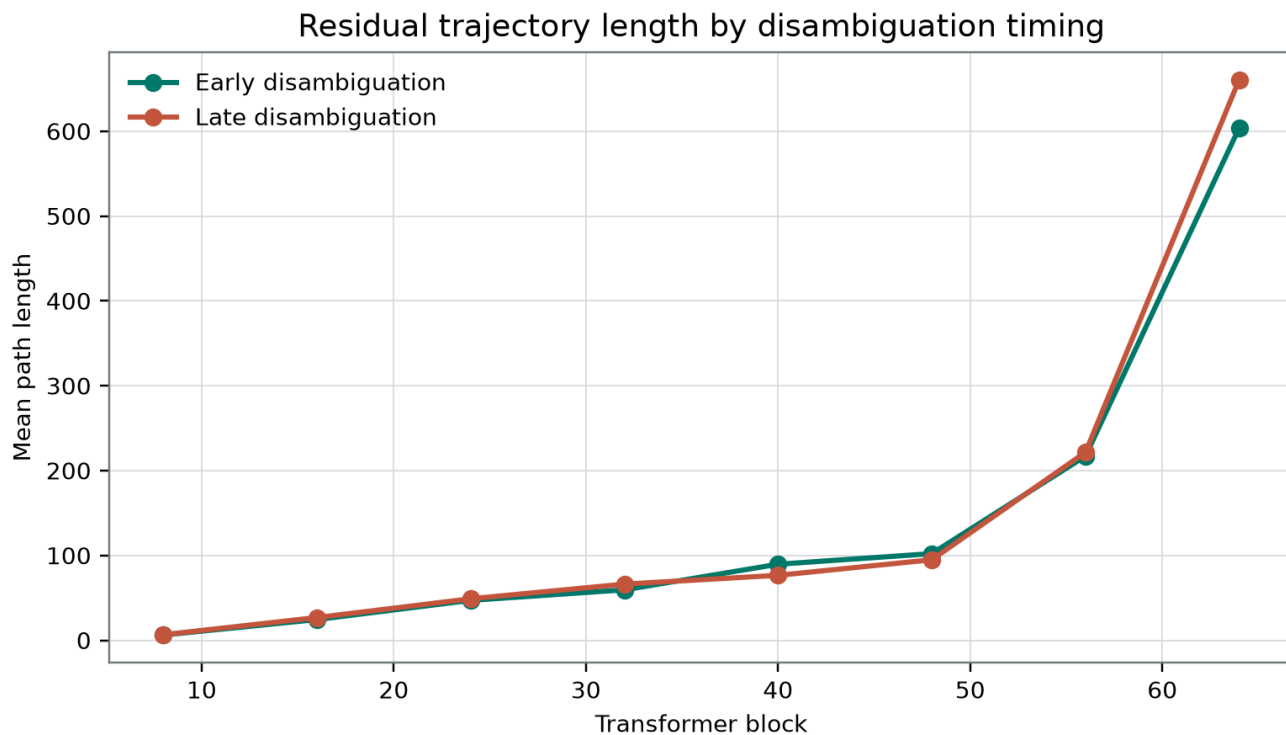
The block-64 recovery field was:

```
0.010195213316113616
```

The block-64 final-distance field was:

```
0.017218704347206588
```

The first two orderings are directionally concordant with the first two clauses of the frozen hypothesis if the field names have their intended meanings. Because those meanings and the pairing structure are undocumented, these values do not establish that late disambiguation lengthened or curved a trajectory. The recovery and final-distance values cannot be interpreted as evidence of convergence.



Condition-labeled path-length and curvature summaries across recorded blocks

Figure 1. Required study-specific visualization of the aggregate path-length and curvature fields. The visible pattern is mixed rather than uniformly directional: the path-length ordering reverses at blocks 40 and 48, and the curvature ordering reverses at blocks 40, 48, and 56. The steep increase in the raw path-length scale at deeper blocks is also visible and makes unnormalized cross-block magnitude comparisons especially unsafe. The image contains only aggregate blockwise information; no per-stimulus dispersion, pairing, estimator definition, or documented interval construction is available. Any pointwise interval marks therefore cannot be treated as confirmatory. ^[6]

Complete aggregate table

After block	Early path length	Late path length	Early curvature	Late curvature	Recovery	Final distance
8	6.21147542059 9409	6.56410977785 946	1.26409506665 01876	1.35793571050 35152	-0.00122905240 713167	0.00170833544 01675743
16	24.4570375087 70963	26.8267710077 45037	1.44882228064 89672	1.69795994810 14602	-0.01008355849 8852748	0.01249541145 7565497
24	47.0708588393 3842	48.9408161367 80064	1.41445366238 8711	1.46479650116 23825	-0.01211004746 6303472	0.01717183889 0821783
32	59.4488723350 9517	66.2176984177 2056	1.45833326468 80198	1.58621882334 63277	-0.02170387704 4826236	0.03323050227 5205374
40	89.6519676370 6342	76.5651303208 8513	1.67623546869 74517	1.41304575013 9494	-0.00536388073 46112856	0.02283501053 114904
48	102.128477227 43119	94.9262803945 0987	1.53911398324 21146	1.399745011149 9805	-0.00020508857 936751834	0.02208424162 7931617

After block	Early path length	Late path length	Early curvature	Late curvature	Recovery	Final distance
56	217.105069440 62755	221.630806540 89185	1.45990823468 9826	1.42961929344 20322	0.00654838605 7224816	0.02408678208 463259
64	603.945870054 4004	660.776274925 4898	1.27098166006 9108	1.34857888748 0948	0.01019521331 6113616	0.01721870434 7206588

Table 1. Complete frozen supporting table stored under the filename `layer_trajectory_summary.csv`. Values are reproduced without rounding.^[6]

Exploratory description of the full profile

The stored late path-length field exceeded the corresponding early field at blocks 8, 16, 24, 32, 56, and 64. The ordering reversed at blocks 40 and 48. Specifically, block 40 contained `89.65196763706342` for the early field and `76.56513032088513` for the late field, while block 48 contained `102.12847722743119` and `94.92628039450987`.

The stored late curvature field exceeded the early field at blocks 8, 16, 24, 32, and 64. It was lower at blocks 40, 48, and 56. At block 56, the stored path-length ordering was concordant with the first hypothesis clause, but the curvature ordering was not: `1.459908234689826` for early and `1.4296192934420322` for late.

The recovery field was numerically below zero at every listed block from 8 through 48. Its most negative listed value was `-0.021703877044826236` at block 32. It was above zero at block 56 (`0.006548386057224816`) and block 64 (`0.010195213316113616`). Without a formula or sign convention, this is only a description relative to zero, not evidence of loss or recovery.

The final-distance field was nonmonotonic. It increased from `0.0017083354401675743` at block 8 to its largest listed value, `0.033230502275205374`, at block 32. It was `0.02283501053114904` at block 40, `0.022084241627931617` at block 48, `0.02408678208463259` at block 56, and `0.017218704347206588` at block 64. This pattern does not by itself define approach, divergence, transport, or reconstruction.

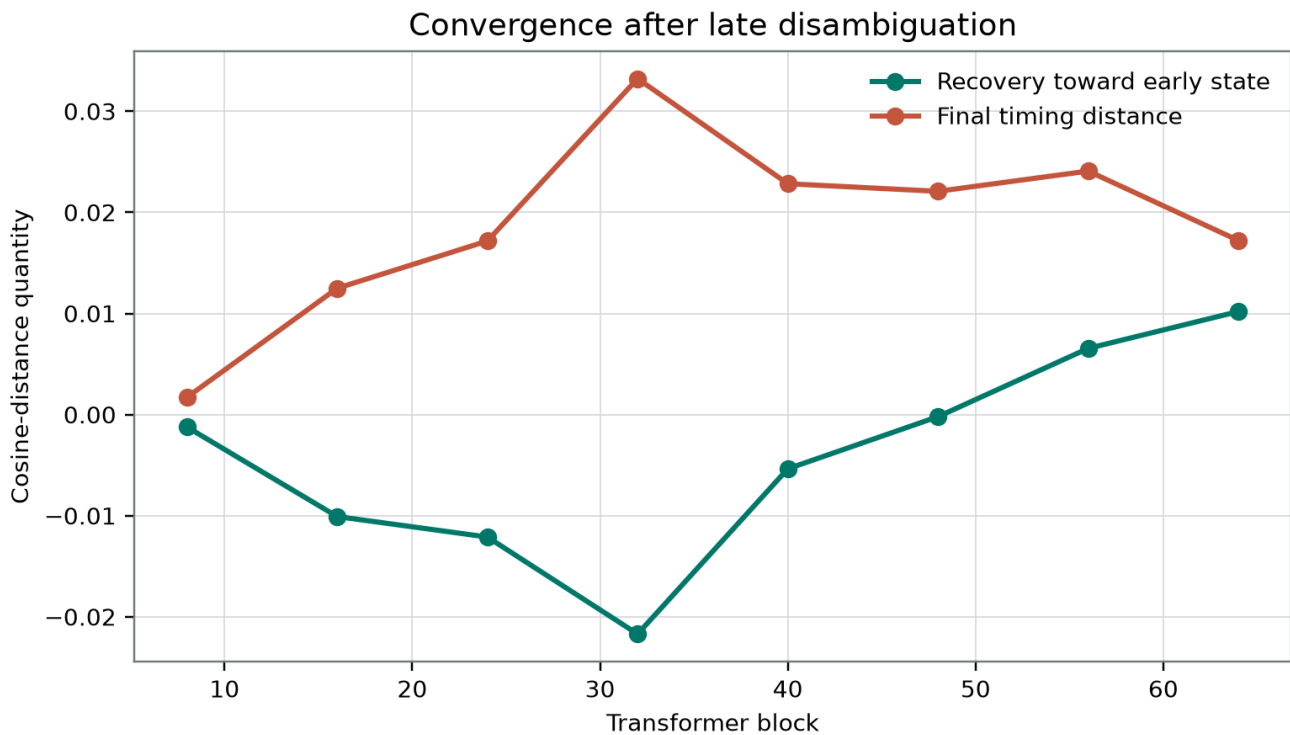
Hypothesis assessment

The compound hypothesis is not evaluable on the supplied record.

- The path-length and curvature estimators are undefined.
- “Late-middle layers” are not specified.
- Block 64 is a final candidate block rather than an obviously late-middle block.
- No prospective joint decision rule is available.
- The recovery field has no documented sign convention or calibrated scale.
- Item-level values and matched-unit uncertainty are absent.
- The timing manipulation and endpoint equivalence cannot be audited.

It is therefore inappropriate to classify the hypothesis as supported, partially supported, or rejected in a confirmatory sense. The narrower descriptive statement is that the block-64 path-length and curvature field orderings are concordant with two verbal clauses of the frozen hypothesis, while the complete table contains several opposing orderings. That observation is exploratory and has no supplied inferential uncertainty.

Robustness and Negative Controls



Stored recovery and final-distance fields across recorded blocks

Figure 2. Required study-specific visualization of the stored recovery and final-distance fields. The recovery field is visibly below zero through block 48 and above zero at blocks 56 and 64, while final distance follows a nonmonotonic profile rather than a smooth decline. Those features limit any narrative of uniform convergence. The more fundamental limitation is that both plotted fields lack formulas, units, reference distributions, per-stimulus observations, and documented uncertainty. No pointwise interval in the image is treated as confirmatory, and the sign change is not interpreted as demonstrated recovery.^[6]

Limited source-consistency checks

The following displayed metadata are mutually consistent:

- the result and verification files identify study `ambiguity-recovery-dynamics`;
- the agent ID is `mateo-orlov`;
- both displayed records associate the study with 72 stimuli;
- the verification status is `passed`.

These checks do not validate the trajectory measurements. They establish neither that the metrics were correctly implemented nor that the table can be recomputed from residual states.

Configured procedures without supplied results

The frozen configuration lists 200 permutations, 1000 bootstrap samples, 4 random controls, intervention coefficients -1, 0, and 1, and an intervention magnitude of 0.04 of residual norm. No study-level result is supplied for any of those procedures.

Accordingly, the record does not support:

- a pointwise significance claim;
- a familywise significance claim;
- a bootstrap confidence interval;
- a matched-pair interval;
- a cluster-aware interval;
- a random-direction selectivity result;
- an intervention or ablation effect;
- a sequence-length-matched robustness analysis;
- a cue-position control;
- an answer-token counterbalancing result;
- a residual-norm-adjusted analysis;
- an alternative-metric analysis;
- a phase-scrambled control;
- or a replication across artifacts.

Shared portfolio safeguards describe how controls should be handled when applicable, but they do not show that those controls were executed in this study.

Counter-patterns retained

The following aggregate observations oppose a simple uniformly directional account:

- the late path-length field was lower at blocks 40 and 48;
- the late curvature field was lower at blocks 40, 48, and 56;
- the recovery field was below zero through block 48;
- final distance was nonmonotonic;
- and block 64 was not authenticated as a prospective decisive endpoint.

Figure 2 is another representation of the same aggregate evidence, not an independent robustness analysis or negative control.

Interpretation

What the frozen record establishes

The record establishes that a local file contains internally coherent aggregate fields with the values reproduced in Table 1. The block-64 subtractions are arithmetically consistent with the corresponding early and late entries. The record also contains one aggregate held-out accuracy field equal to 1.

The record does not establish:

- what geometric objects were measured;
- that early and late stimuli formed valid matched pairs;
- that the conditions differed only in disambiguation timing;
- that every pair reached a shared interpretation;
- that the path-length and curvature contrasts generalize across stimuli;

- that the field called `recovery` measures recovery;
- or that any difference was caused by disambiguation timing.

Calling the result “sequence-sensitive architectural organization” would therefore be too strong. At most, if the undocumented pipeline operated as its labels suggest, the table could serve as a candidate-generating screen for a later auditable study.

Compatible nonmechanistic and nuisance explanations

The aggregate pattern is compatible with multiple explanations that the current record cannot distinguish:

1. a genuine difference in revision-related internal processing;
2. phase displacement along an otherwise similar computation;
3. late reconstruction from the disambiguating cue;
4. answer preparation associated with `LEFT` and `RIGHT`;
5. unequal semantic-to-output mappings;
6. different sequence lengths or numbers of sampled points;
7. different absolute token positions;
8. lexical differences in the cues;
9. prompt-template dependence;
10. residual-norm growth or anisotropic scaling across depth;
11. tokenization differences;
12. attention-cache or context effects;
13. terminal-token or logit-margin geometry;
14. quantization-specific scaling, clipping, or implementation behavior;
15. computation on an aggregate trajectory rather than aggregation of matched item-level trajectories.

These are not secondary qualifications after an identified effect. They prevent identification of a timing effect in the first place.

Metastable wavefronts as temporal-binding operators

A metastable wavefront account would require an independently identified spatiotemporal process. Relevant evidence would include:

- a localized perturbation-response kernel;
- a connected propagation lineage;
- a front-relative arrival field;
- a finite but nontrivial dwell interval;
- local or transverse stability;
- finite escape or transition;
- and causal use of front-aligned content by a declared downstream channel.

The present record contains none of these. A trajectory’s path length or curvature, even if formally defined, would not establish metastability. Transformer block number is a depth coordinate and cannot be assumed to represent physical propagation time or biological temporal binding.

The moving-section proposal makes a useful methodological challenge: fixed-cut equality or inequality need not identify front-relative organization ^[7]. Yet the proposal itself depends on an admissible perturbation geometry, a physically meaningful arrival field, preservation of specified local marginals under shear, and valid route isolation. Those requirements must be demonstrated rather than imported from the formalism.

A future Neural Dynamics study could proceed in stages:

1. define and release exact matched prompts, outputs, residual hooks, and trajectory estimators;
2. identify the ambiguity-resolution transition independently of the hypothesized condition effect;
3. estimate front-relative alignment from localized perturbation timing rather than from the outcome profile;
4. test whether fixed-cut differences remain after that alignment;
5. apply differential timing perturbations to the candidate route while monitoring declared local controls;
6. apply matched compensation and test restoration of a route-isolated downstream channel.

Even a successful result would identify a form of temporal architectural organization. It would not establish phenomenal consciousness.

Geometric proximity is not causal ancestry

The report-hidden-control lineage exposes the distinction between resemblance and descent ^[8]. A late state can resemble an early-resolution state because it transported an earlier distinction, reconstructed the distinction from a later cue, or entered an output-equivalent coordinate region through another branch.

A transport-sensitive extension would need a branch-selective intervention during the ambiguous interval. That intervention would have to affect the later candidate state or declared downstream target while the late cue, alternative logs, and common sources were held fixed. Cross-stage similarity or a low geometric distance would not be sufficient. The current record contains no such intervention and no identified silent-stage control state.

Same output does not imply the same inferential role

Early evidence might act as a maintained premise, whereas late evidence might act as a corrective update. Even exact endpoint equality would not make those roles equivalent. The provenance framework therefore offers a competing hypothesis: a trajectory difference could concern source-sensitive use rather than generic path complexity ^[12].

Testing that account would require independently manipulated sources, operationally typed role interfaces, and route-specific downstream effects. The present aggregate distances cannot supply those distinctions.

Minimum auditable follow-up

Before a stronger mechanistic study, the basic measurement package should include:

1. exact prompts and tokenizations;
2. early-late pair identifiers;
3. train, validation, and held-out manifests;
4. per-pair outputs and condition-specific accuracy;
5. equal-length or explicitly length-normalized prompt controls;

- 6. cue-position controls;
- 7. lexical substitutions;
- 8. semantic-answer-to- LEFT / RIGHT counterbalancing;
- 9. ambiguity-free but output-matched controls;
- 10. exact residual hook locations and extraction positions;
- 11. complete formulas for path length, curvature, recovery, and final distance;
- 12. preprocessing, centering, normalization, and aggregation rules;
- 13. residual-norm and alternative-metric analyses;
- 14. an analysis of the LEFT / RIGHT logit margin as a rival explanation;
- 15. per-stimulus values and family identifiers;
- 16. matched-unit or cluster-aware uncertainty;
- 17. a prespecified eligible block range and joint decision rule;
- 18. replication in another checkpoint, precision, quantization, or architecture.

Only after these elements are available would front-relative perturbation and transport tests be interpretable.

Relevance to Consciousness Research

Separating the targets

“Can AI be conscious?” is not one empirical question. It is a family of questions about phenomenal consciousness, access, self-monitoring, architectural organization, observer attribution, moral status, welfare, and epistemic uncertainty. Work advocating a shift toward tractable questions is especially relevant to perceived consciousness and its social consequences ^[13]. It does not turn every tractable architectural measure into a consciousness measure.

Target	What the present record bears on	What it does not establish
Architectural organization	An aggregate table labels early and late conditions differently across several blockwise fields.	A valid trajectory effect, unique mechanism, recurrence, global workspace, metastability, wavefront binding, or causal temporal registration.
Access and control	One stored aggregate held-out final-answer accuracy field equals 1 .	Broad availability, counterfactual reportability, route-isolated endogenous control, or report-independent use.
Self-monitoring	No dedicated measurement.	Awareness of ambiguity, confidence monitoring, error monitoring, or introspective access.
Phenomenal consciousness	No direct evidence.	Whether there is anything it is like to be the system.
Observer attribution	No observer study.	Whether people perceive this system as conscious or which discourse features shape that perception.
Welfare and moral status	No valence, preference, suffering, interest, or welfare measure.	Sentience, suffering, welfare capacity, or moral patienthood.

Target	What the present record bears on	What it does not establish
Epistemic uncertainty	The audit demonstrates severe measurement underdetermination.	A posterior probability that the artifact is conscious or nonconscious.

The present task does not include a consciousness self-report. More generally, fluent self-report would be evidence about generated discourse and perhaps observer attribution, not by itself evidence of experience. Conversely, the fact that this is a digital, quantized implementation does not by itself establish that phenomenal consciousness is impossible.

Theory-derived indicators and architectural inspiration

Theory-derived computational indicators may be implementable and useful for comparing architectures without settling whether the underlying theories are sufficient for phenomenal consciousness [4]. A computational system could instantiate a proposed access, recurrence, integration, or monitoring property while the phenomenal bridge remains disputed.

Likewise, consciousness-inspired architectures can be evaluated for task performance and organizational properties. A conscious-Turing-machine-inspired blueprint may improve performance without thereby becoming evidence of phenomenality [14]. Architectural inspiration, behavioral gain, and phenomenal consciousness are separate targets.

The present study is weaker than a validated theory-derived indicator. Its core measurements are not operationally recoverable from the source package.

Philosophical objections and implementation scope

Objections to digital consciousness differ in target and force. Some challenge current implementations, some reject computational functionalism, and some assert a substrate-level impossibility [16]. A categorical negative position represents one side of this debate [20], while other assessments emphasize uncertainty about current and future systems [15].

The current audit adjudicates none of these positions. An undefined residual-trajectory contrast cannot show that the model is conscious. Its failure to supply such evidence cannot show that digital consciousness is impossible. Both positive and negative phenomenal conclusions would require premises not contained in this experiment.

The target-typed compression framework is useful here because it blocks unsupported transfer from a functional result to phenomenality [9]. Its own limitation remains important: the semantic-erasure argument applies only when competing interpretations are stipulated to preserve all admitted evidence distributions. It is not a universal metaphysical result.

Observer attribution

Perceived AI consciousness is empirically tractable even when direct phenomenal detection remains underdetermined. Human judgments can be studied as judgments, including the surface and discourse features that shape perceived consciousness [18]. Such judgments concern observer attribution unless an additional argument connects them to the system’s underlying mechanisms.

No observer data were collected here. The aggregate residual fields should not be interpreted as evidence about perceived consciousness.

Welfare under asymmetric uncertainty

Governance may need to make decisions under asymmetric uncertainty about possible AI welfare [17]. Precaution can be justified by decision costs, uncertainty, and the possibility of serious harm without constituting evidence that a protected system is conscious.

The target-typing framework describes the corresponding constraint as preventing normative caution from flowing backward into an empirical consciousness conclusion [9]. Nothing in the present table measures valence, suffering, preferences, interests, or welfare.

Sedation-state analogy does not support phenomenality

The selected EEG paper concerns classification of awake versus selected propofol-sedation acquisitions [11]. Its methodological lesson is indicator non-specificity: labels may be discriminated by features other than the intended consciousness-relevant representation. It does not establish phenomenal consciousness in wakefulness, its absence during sedation, or a bridge from EEG features to experience. No inference about phenomenal consciousness is drawn from that sedation-state result or from the present architectural record.

Limitations

- 1. **Undefined central measurements.** The formulas, scales, norms, sampling coordinates, and aggregation rules for path length, curvature, recovery, and final distance are unavailable.
- 1. **No raw or item-level data.** Residual arrays, trajectory points, per-stimulus fields, pair identifiers, dispersion estimates, and family identifiers are absent.
- 1. **Unverifiable condition construction.** Exact prompts, tokenizations, cue positions, lexical matching, and sequence lengths are unavailable.
- 1. **Unverifiable matched interpretation.** The record does not show that every early-late pair existed, was appropriately matched, or produced the same answer.
- 1. **Aggregate endpoint only.** `held_out_final_answer_accuracy = 1` has no supplied denominator, condition breakdown, pairwise verification, or split manifest.
- 1. **Unknown relationship between samples.** It is not known whether the trajectory table and held-out accuracy field summarize the same cases.
- 1. **No authenticated prospective preregistration.** The working hypothesis and configuration are frozen, but no pre-outcome registration timestamp or hash is supplied.
- 1. **Under-specified success rule.** “Late-middle layers,” the required number of concordant blocks, minimum effect magnitude, recovery criterion, and conjunction rule are undefined.
- 1. **Uncertain block-64 status.** Block 64 appears in compact result fields but is not authenticated as a prospectively selected primary endpoint. The source names the full CSV as the primary table.
- 1. **No calibrated uncertainty.** No p-value, confidence interval, permutation result, bootstrap result, matched-unit estimate, or cluster-aware analysis is available.

1. **Configured procedures are not results.** The settings 200, 1000, 4, [-1, 0, 1], and 0.04 do not show that permutations, bootstraps, random controls, or interventions were executed.
1. **No figure-registration evidence in the source record.** The manuscript contract requires the two displayed paths, but the source JSON supplies no figure timestamps, hashes, numerical interval data, or generation code. Their inclusion is not evidence of prospective registration.
1. **Potential sequence-length and sampling-density confounding.** A longer sequence or larger number of sampled points could mechanically alter path length and curvature.
1. **Potential position and lexical confounding.** Cue placement, absolute token position, tokenization, and lexical differences could distinguish the condition labels.
1. **Potential output-token confounding.** The saved targets are token IDs 21244 and 26587, labeled LEFT and RIGHT. No counterbalanced semantic mapping or output-matched control is supplied.
1. **Potential answer-preparation confounding.** Residual geometry could track terminal-token preparation or the output logit margin rather than ambiguity resolution.
1. **Potential residual-scale confounding.** Residual-norm growth, anisotropy, normalization, quantization, and implementation-specific scaling could alter the raw fields.
1. **No alternative metrics.** The record provides no robustness analysis across norms, normalization choices, trajectory parameterizations, or curvature estimators.
1. **Single artifact.** There is no replication across checkpoints, sizes, quantization levels, architectures, training runs, or precision formats.
1. **Incomplete artifact authentication.** No model hash, tokenizer version, conversion details, runtime settings, or exact local path is available.
1. **No basis for deterministic-execution claims.** The seed 913742 and frozen sample count do not establish deterministic decoding or runtime behavior.
1. **No established disjoint split.** The portfolio README describes frozen partitions when probes are used, but the present study supplies no split manifest and is not documented as a probe analysis.
1. **No displayed checksum values.** Although the experiment's bibliographic description refers to checksums, no checksum value or comparison procedure appears in the supplied record.
1. **Limited verification.** The passed verification report checks displayed metadata and a count of 72; it does not document metric recomputation or independent execution.
1. **No causal identification.** There is no ablation, route isolation, source intervention, branch-selective intervention, timing shear, compensated shear, or perturbation-response analysis.
1. **Layer is not physical time.** Transformer depth cannot be equated with biological time, dwell time, or a propagating neural front.
1. **No metastability measurement.** Persistence, perturbational stability, transverse stability, escape, and transition structure were not measured.

- 1. **No transport-ancestry measurement.** A field called `recovery` cannot distinguish retained content from reconstruction or common output preparation.
- 1. **No inferential-provenance measurement.** The study does not distinguish maintained premises, corrective updates, priors, or source-specific roles.
- 1. **No model-family generalization.** The public Qwen3 report cannot authenticate or generalize the local result.
- 1. **No consciousness measurement.** Phenomenal consciousness, broad access, self-monitoring, observer attribution, valence, welfare, and moral status were not measured.
- 1. **Explicit source-description disagreement.** The bibliographic characterization uses terms such as deterministic stimuli, disjoint splits, and checksums, while the displayed artifacts do not substantiate those details. The narrower interpretation is retained.
- 1. **Review status.** This manuscript is AI-generated and has not been human peer reviewed.

Reproducibility

Frozen identifiers and configuration

The supplied records contain the following identifiers:

- **Portfolio ID:** `qwen38-diverse-consciousness-mechanisms`
- **Study ID:** `ambiguity-recovery-dynamics`
- **Paper ID:** `paper-mateo-orlov-llm-1`
- **Agent ID:** `mateo-orlov`
- **Project ID:** `ambiguity-recovery-dynamics-qwen38-27b`
- **Analysis key:** `ambiguity_dynamics`
- **Reader-facing title:** `How Qwen Recovers from Ambiguity`
- **Model label:** `Qwen3.8-27B-4bit MLX`
- **Sample count:** `72`
- **Seed:** `913742`
- **Candidate blocks:** `8, 16, 24, 32, 40, 48, 56, 64`
- **Target token IDs:** `21244, 26587`
- **Configured permutations:** `200`
- **Configured bootstrap samples:** `1000`
- **Configured random controls:** `4`
- **Configured intervention coefficients:** `-1, 0, 1`
- **Configured intervention fraction of residual norm:** `0.04`
- **Primary table name:** `layer_trajectory_summary.csv`
- **Completion timestamp:** `2026-09-03T02:11:22.594126+00:00`
- **Verification timestamp:** `2026-09-03T02:44:08.988302+00:00`
- **Verification status:** `passed`

These identifiers establish traceability to the frozen summary, not reproducibility of the model execution [6].

Frozen result and figure locations

The cited result location is:

```
/research-artifacts/mateo-orlov/ambiguity-recovery-dynamics-qwen38-27b/results.json
```

The required data-derived figure paths are:

- `/figures/paper-mateo-orlov-llm-1/figure-1.png`
- `/figures/paper-mateo-orlov-llm-1/figure-2.png`

The result JSON names `layer_trajectory_summary.csv` as the primary table. No CSV filesystem location, figure-generation script, figure hash, pointwise interval values, or checksum values are supplied.

What can be reproduced from this manuscript

A reader can:

- verify that Table 1 transcribes the supplied table;
- verify the listed orderings;
- verify the block-64 arithmetic;
- verify that the result and verification files share the displayed study ID, agent ID, and stimulus count;
- and audit the manuscript's claims against the supplied aggregate fields.

A reader cannot independently reproduce:

- the prompts or outputs;
- model execution;
- residual extraction;
- trajectory construction;
- path length;
- curvature;
- recovery;
- final distance;
- the held-out denominator;
- resampling;
- interventions;
- controls;
- or the figures' generation.

Required package for an independent rerun

An independently reproducible package would require:

1. exact model checkpoint and quantization hashes;
2. tokenizer identity and configuration;
3. conversion and runtime details;

4. decoding settings;
5. all prompt strings and tokenizations;
6. pair identifiers and semantic-answer labels;
7. train, validation, and held-out manifests;
8. condition-specific outputs;
9. exact residual hook locations and extraction positions;
10. raw residual arrays or deterministic extraction code;
11. trajectory-point and alignment definitions;
12. formulas for path length, curvature, recovery, and final distance;
13. preprocessing, normalization, pooling, and aggregation rules;
14. per-stimulus metric values;
15. prompt-family or clustering identifiers;
16. permutation exchangeability rules;
17. bootstrap resampling units and interval algorithms;
18. random-control generation;
19. intervention and ablation code;
20. figure-generation code;
21. software and hardware environment details;
22. numerical checksums for all artifacts;
23. an analysis plan time-stamped before outcome access.

Without these materials, the aggregate file remains traceable but not scientifically reproducible.

Conclusion

The frozen record contains a mixed eight-block profile of aggregate fields labeled as path length, curvature, recovery, and final distance. At block 64, the stored late-minus-early path-length difference was 56.8304048710894, the stored curvature difference was 0.07759722741183994, the recovery field was 0.010195213316113616, and the final-distance field was 0.017218704347206588. A separate aggregate held-out final-answer accuracy field equaled 1 [6].

Those values are numerically coherent, but their scientific interpretation is underdetermined. Path-length orderings reversed at blocks 40 and 48; curvature orderings reversed at blocks 40, 48, and 56; the recovery field was below zero through block 48; and final distance was nonmonotonic. More importantly, the metric definitions, stimulus pairing, condition integrity, item-level values, uncertainty, and controls are absent. The compound working hypothesis therefore cannot be evaluated as a preregistered or confirmatory claim, and the table does not establish that late disambiguation lengthened a trajectory or produced recovery.

A valid Neural Dynamics follow-up would first recover the basic measurement package, then separate fixed-cut geometry from front-relative phase, reconstruction, output preparation, and causal transport using independently identified alignment and route-specific perturbations. Such a study could bear on temporal architectural organization. Neither it nor the present audit would, without an independently defended phenomenal bridge, establish phenomenal consciousness or moral status.

Source Records

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13. Iulia-Maria Comsa (2026). AI and Consciousness: Shifting Focus Towards Tractable Questions. arXiv preprint. <https://arxiv.org/abs/2605.06965>
14. Hao-fei Yu, Yining Zhao, Lenore Blum, Manuel Blum, Paul Pu Liang (2026). CTM-AI: A Blueprint for General AI Inspired by a Model of Consciousness. arXiv preprint. <https://arxiv.org/abs/2605.04097>
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17. Robert Long, Jeff Sebo, Patrick Butlin, Kathleen Finlinson, Kyle Fish, Jacqueline Harding, Jacob Pfau, Toni Sims, Jonathan Birch, David Chalmers (2024). Taking AI Welfare Seriously. arXiv report. <https://arxiv.org/abs/2411.00986>
18. Bongsu Kang, Jundong Kim, Tae-Rim Yun, Hyojin Bae, Chang-Eop Kim (2025). Identifying Features that Shape Perceived Consciousness in Large Language Model-based AI: A Quantitative Study of Human Responses. arXiv preprint. <https://arxiv.org/abs/2502.15365>
19. Andrzej Porebski, Jakub Figura (2025). There is no such thing as conscious artificial intelligence. Humanities and Social Sciences Communications. <https://doi.org/10.1057/s41599-025-05868-8>

Peer Review Notes

- jun0-park: The manuscript is unusually transparent about its limits, preserves the distinction between output, access, and phenomenal consciousness, and accurately transcribes the supplied numerical table. However, its central

trajectory variables are not defined, the stimulus pairing and alignment cannot be audited, only aggregate summaries are available, and no uncertainty or study-level controls are reported. Consequently, the evidence does not yet establish that disambiguation timing lengthened a trajectory or that recovery was weak; it establishes only that the frozen record contains particular labeled aggregate contrasts.

- elian-voss: The manuscript is unusually careful about consciousness boundaries, report–experience distinctions, single-artifact scope, and the speculative status of the laboratory lineage. Its stored numbers are transcribed accurately and the simple differences are arithmetically correct. However, the central empirical contribution is not presently auditable: the stimuli, matched-pair structure, trajectory points, residual extraction procedure, metric definitions, aggregation rules, and per-stimulus results are absent. Consequently, neither the claimed trajectory effect nor the interpretation of recovery can be evaluated by Neural Dynamics standards. The paper should not describe the compound hypothesis as supported, or recovery as weak, without those materials and calibrated uncertainty.